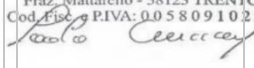


UDROŻNIENIE ŁÓDZKIEGO WĘZŁA KOLEJOWEGO (TEN-T), ETAP II, ODCINEK ŁÓDŹ FABRYCZNA - ŁÓDŹ KALISKA/ŁÓDŹ ŻABIENIEC – PROJEKT POIIS 5.1-15

TBM 13.08 M - CONSTRUCTION FOLLOW-UP

IDENTIFICATION TABLE

Client	PBDiM - Energopol
Project	Udrożnienie Łódzkiego Węzła Kolejowego (TEN-T), Etap II, Odcinek Łódź Fabryczna - Łódź Kaliska/Łódź Żabieniec – Projekt POIIS 5.1-15
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1. INTRODUCTION

S-1222 TBM EPB machine with 13.08 m excavation diameter is currently mining Axis 23, in the section between Polesie Station and Śródmieście Station. TBM has restarted from chainage 1+962.

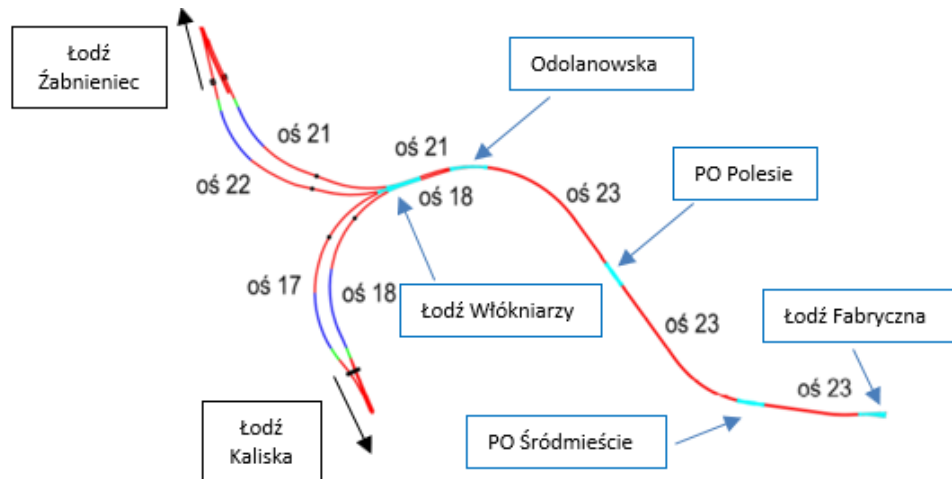


Figure 1-1 Overview of main structural elements of the Project

This daily follow-up report aims to give a general overview of S-1222 TBM drive during 27/08/2024. It includes EPB face pressure, belt scale parameters, thrust and torque values, primary grout parameters, guidance, as well as monitoring follow-up data.

According to VMT system, TBM is at Chainage 1+703.32 after the erection of ring n. 222; which corresponds to a total excavation length of 263.15 m from the starting chainage.

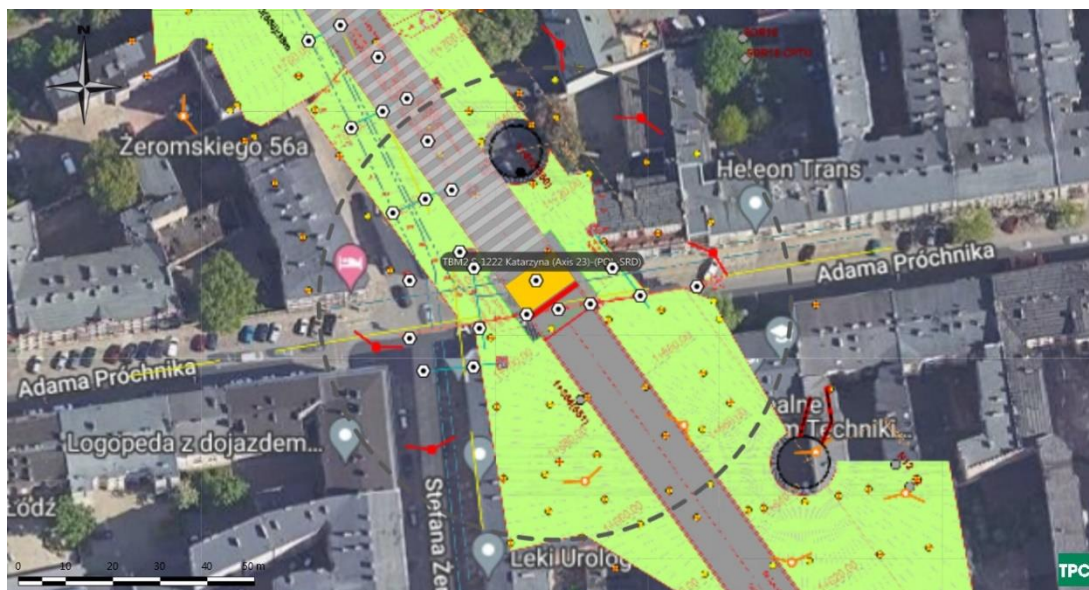


Figure 1-2 Plan view of TBM Position on 28/08/2024

2. FACE PRESSURES AND EXCAVATION VOLUMES

Average EPB pressure at the crown is at about 1.7 bar. The value is within the attention values. At the end of ring 221, a face pressure drop to 1.4 bar has occurred. Instable values of face pressure are present also at the end of ring excavation 222.

Figure 2-1 shows EPB sensors.

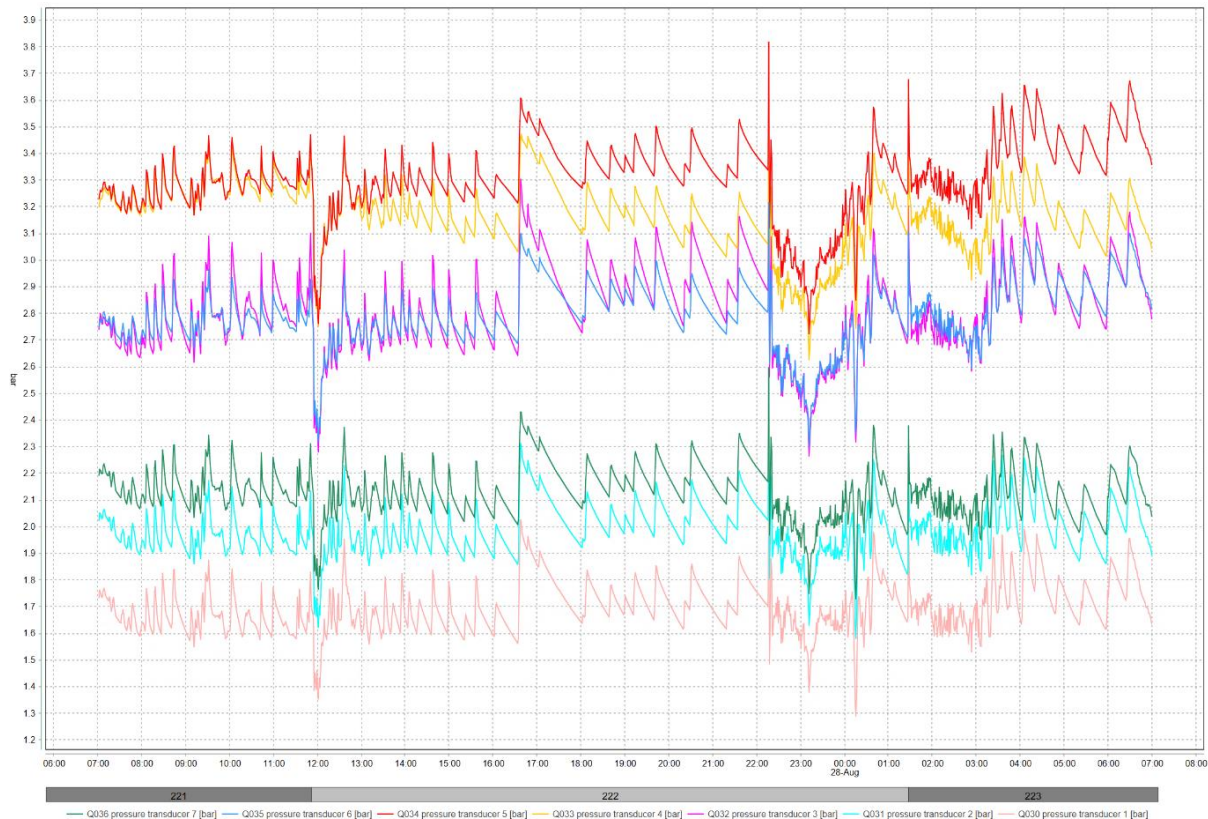


Figure 2-1 EPB pressure sensor

The delta between pressure transducer 1 and 2 has an average value of 0.29 bar indicating that the bulk chamber is correctly filled with material. At the end of ring 222 a slight drop of delta value to 0.17 bar has occurred.

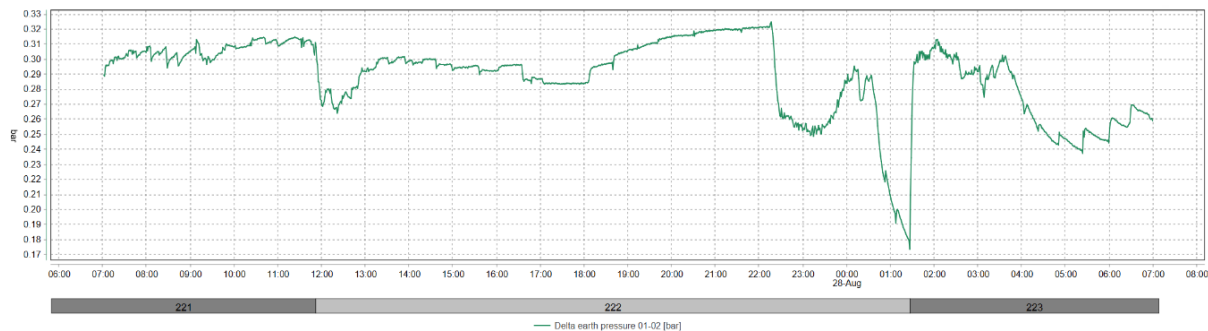


Figure 2-2 Delta between pressure sensors 1 and 2

Belt scale n. 2 show values much lower than n. 1. The results of Belt scale n. 2 seems to be affected by a constant error and are therefore considered not representative. The belt scale value is a raw indication of the excavated material quantity.

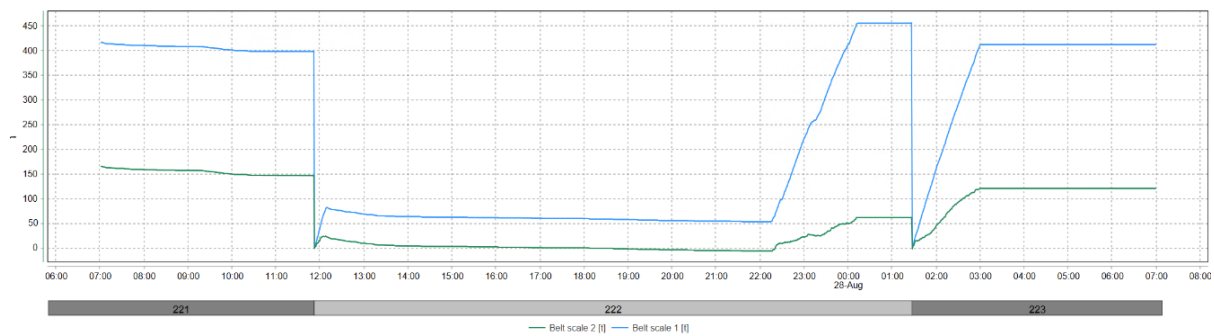


Figure 2-3 Belt scales

Table 2-1 Belt scale values at the end of each advance

Ring	Belt scale 1 (t)	Belt scale 2 (t)
221	421	172
222	457	63

3. THRUST AND CUTTERHEAD TORQUE

During excavation, the average torque is of 4.3 MNm.

During advancement, average advance rate is of 16 mm/min.

During excavation, thrust force shows an average value of 77.5 MN and a maximum of 86.6 MN.

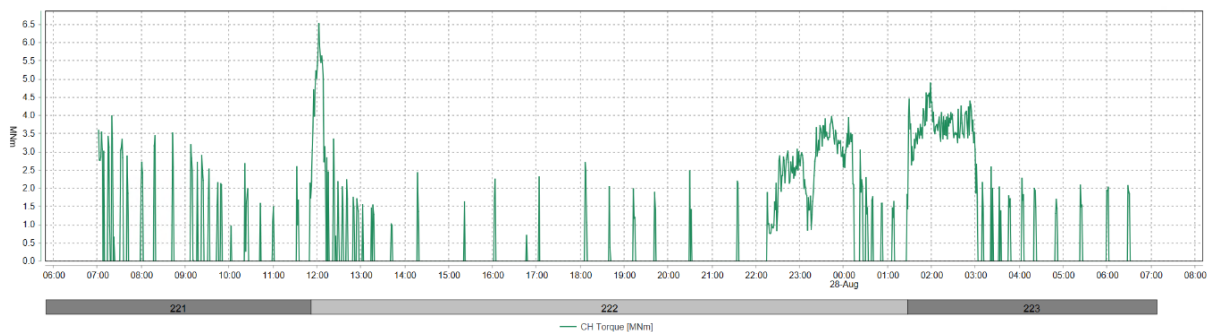


Figure 3-1 CH Torque

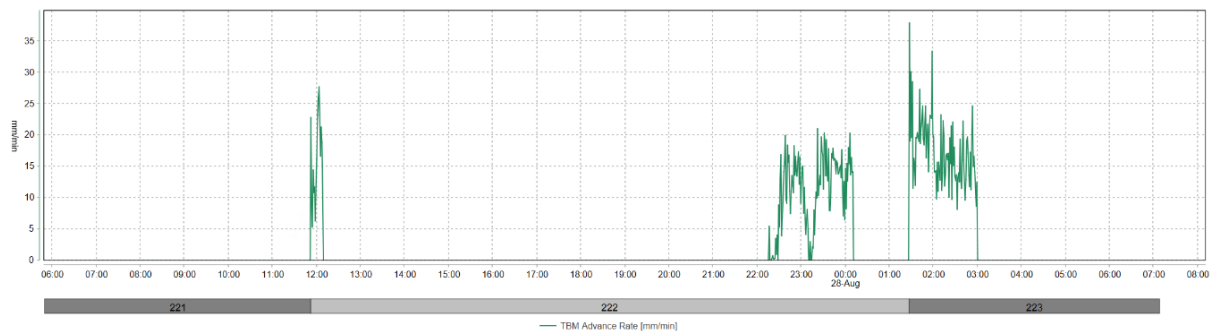


Figure 3-2 Advance rate

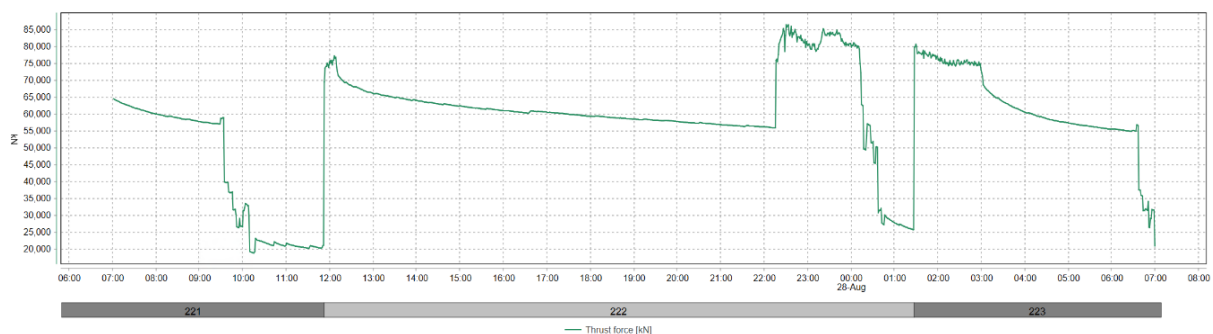


Figure 3-3 Thrust force

4. GROUT INJECTION

Grout injection has a volume above the minimum theoretical annular volume $\pm 10\%$ of 13.54 m^3 . Ration B/A is at 7%.

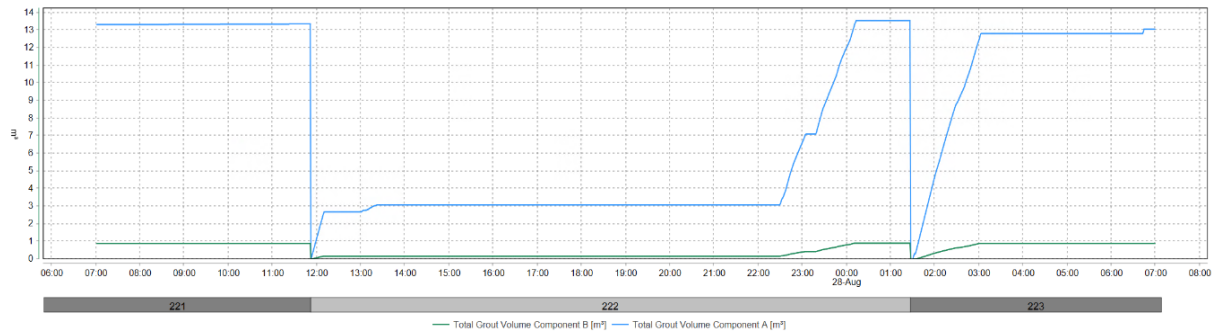


Figure 4-1 Grout volumes

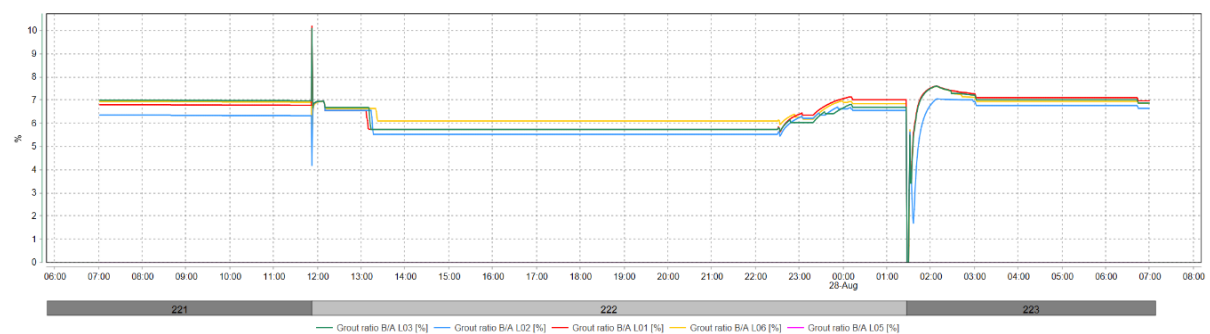


Figure 4-2 Volume ratio Components B/A

Table 4-1 Grout volumes per ring

Ring	Component A (m ³)	Component B (m ³)	B/A (%)
221	13.4	0.9	7
222	13.6	0.9	7

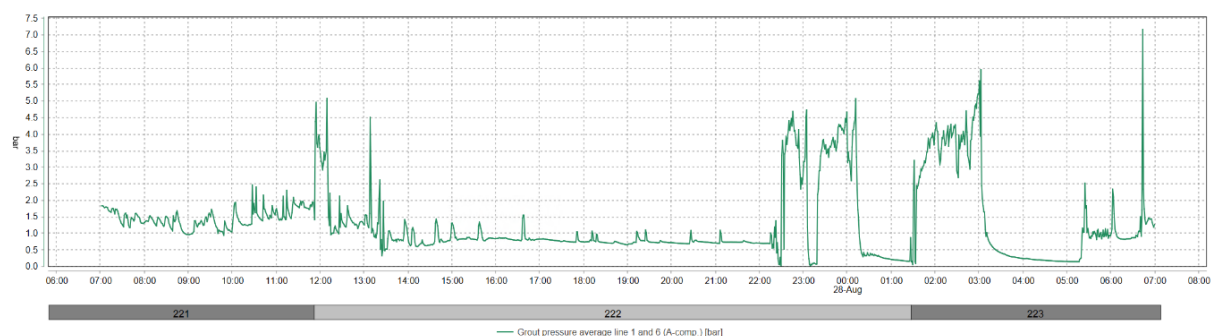


Figure 4-3 Average injection pressure line 1 and 6

5. GUIDANCE

The TBM vertical deviation is of about 10.52 mm on the front edge of the shield and the horizontal deviation of about 18.31 mm (Figure 5-1). Maximum radial deviation is of 21.84 mm.

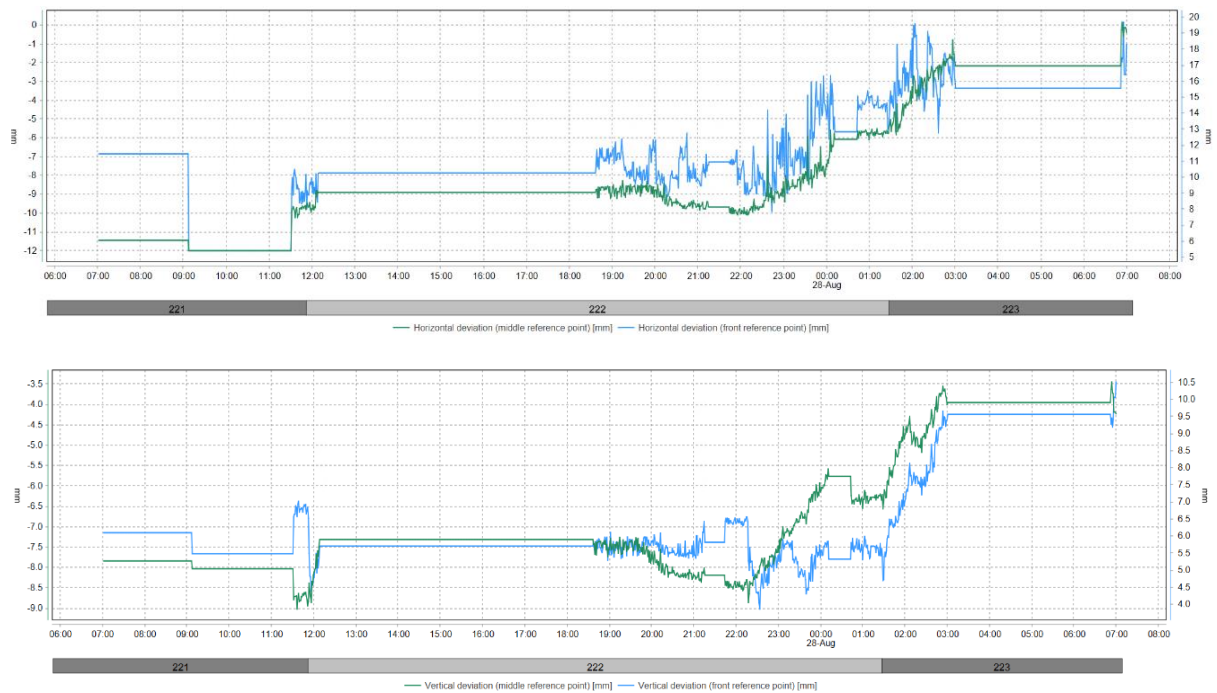


Figure 5-1 Machine deviation from the alignment

6. MONITORING

According to Geoscope, settlements of the buildings above the tunnel are kept within the limits.

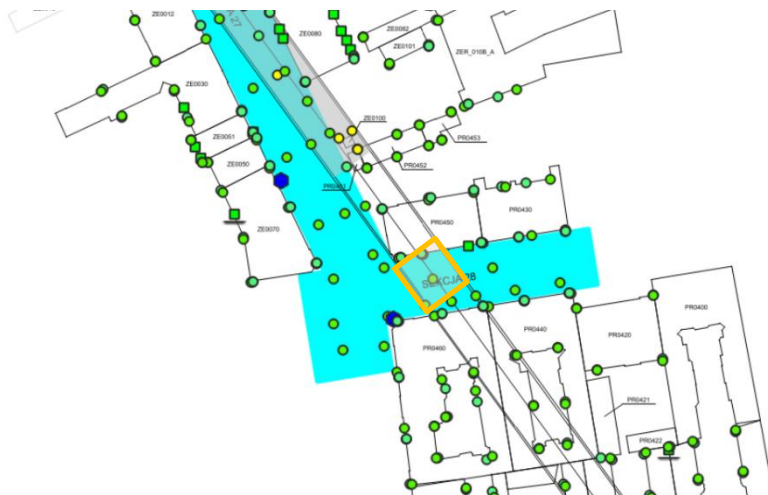


Figure 6-1 Monitoring points

The monitoring points on Zeromskiego street are stable.

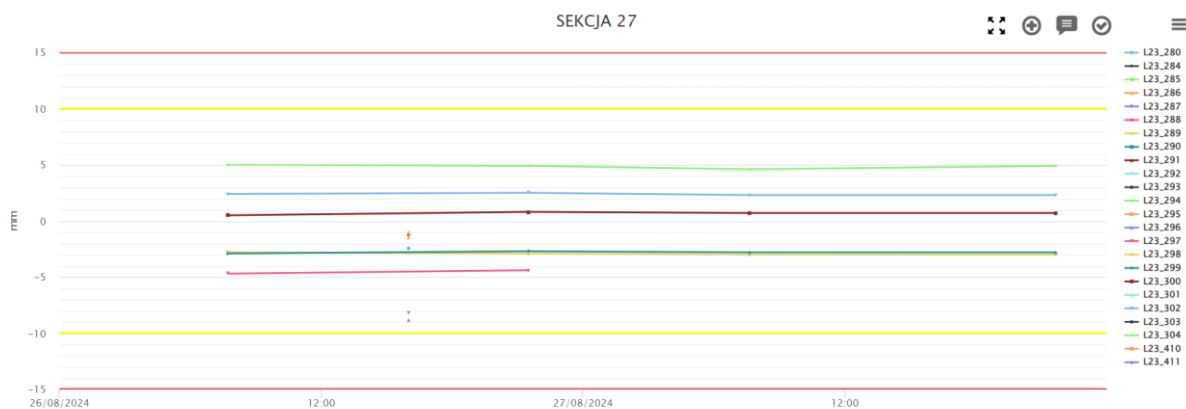


Figure 6-2 Settlement of Section on Zeromskiego street

The monitoring points on Prochnika street show a settlement values of about 3 mm in 48 hours, particularly in point L23_308, L23_313 and L23_314, that are the point closest to tunnel axis.

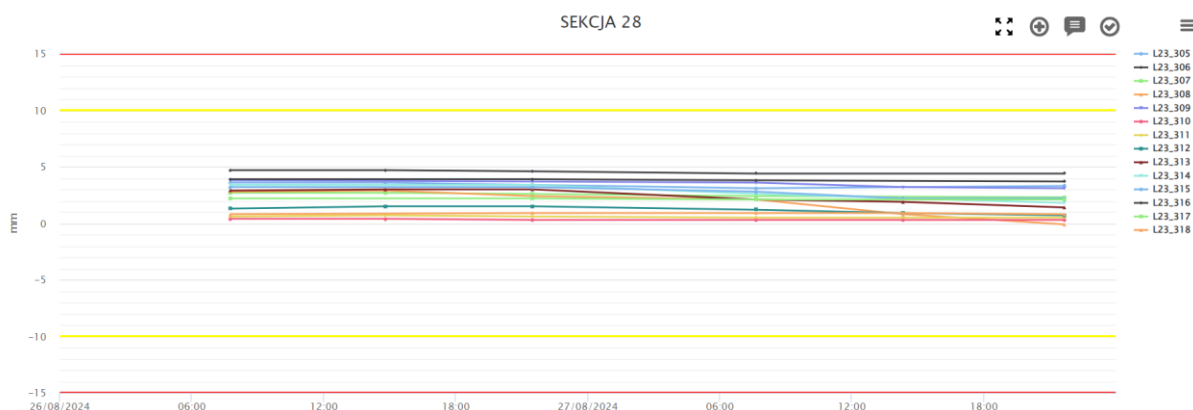


Figure 6-3 Settlement of Section on Prochnika street

In building PR0452, monitoring points are stable except for point 02, that shows small downward trend.

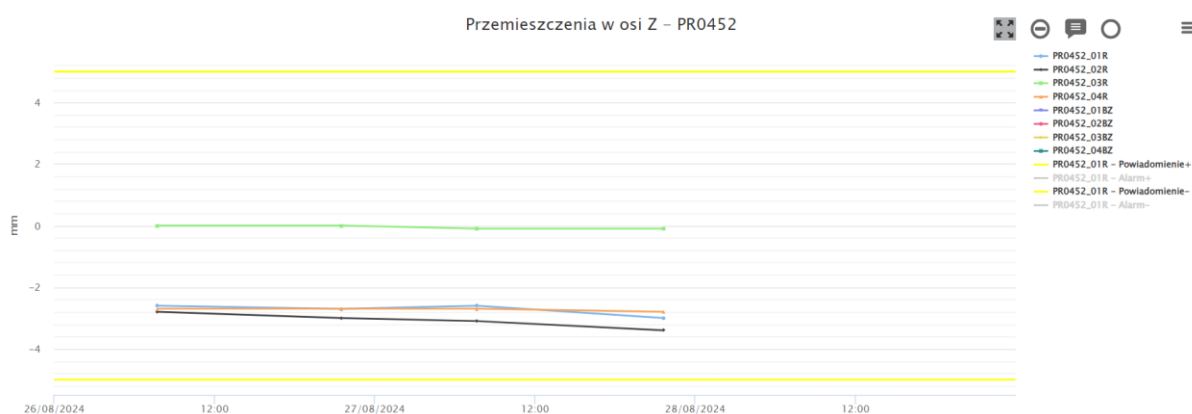


Figure 6-4 Settlement of building PR0452

On PR0450, monitoring points are stable in the last 24 hours.

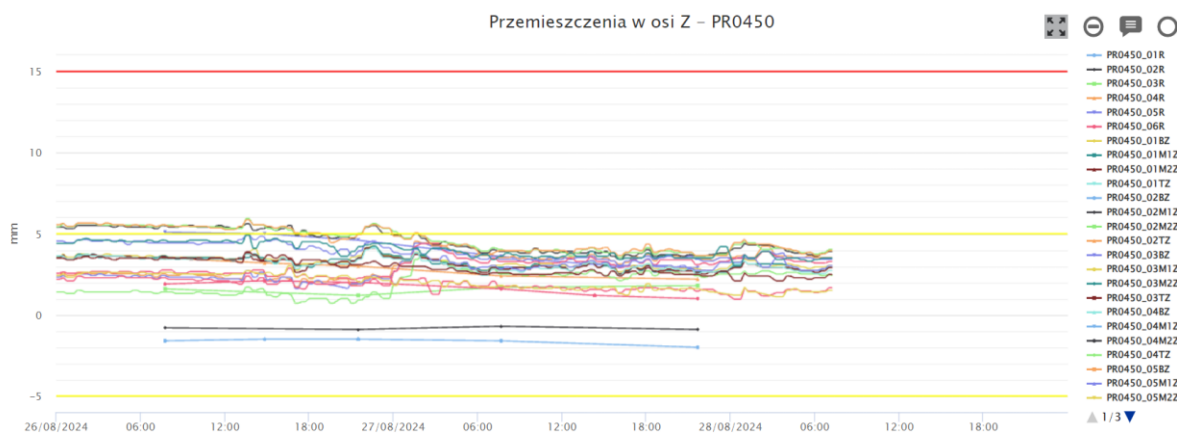


Figure 6-5 Settlement of building PR0450

On ZE0070, monitoring points values are stable.

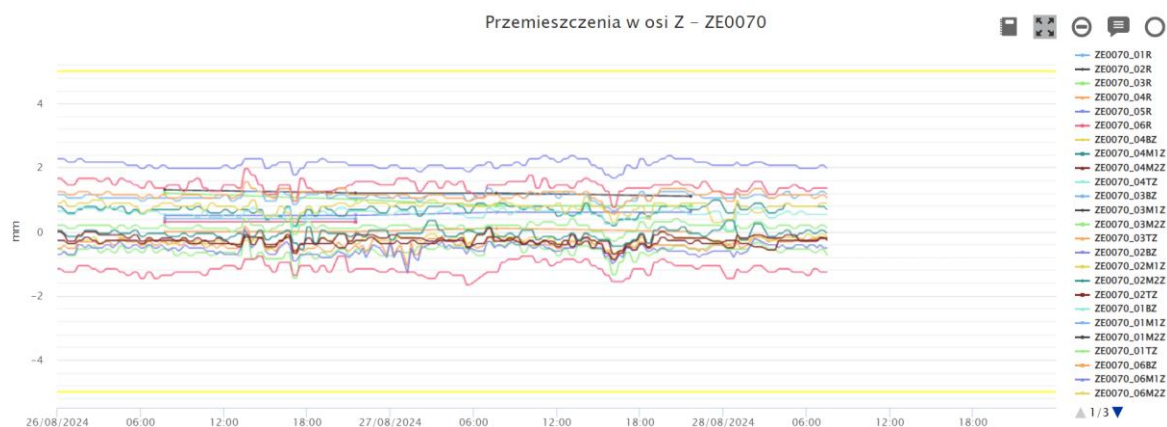


Figure 6-6 Settlement of building ZE0070

On PR0430, monitoring points are stable.

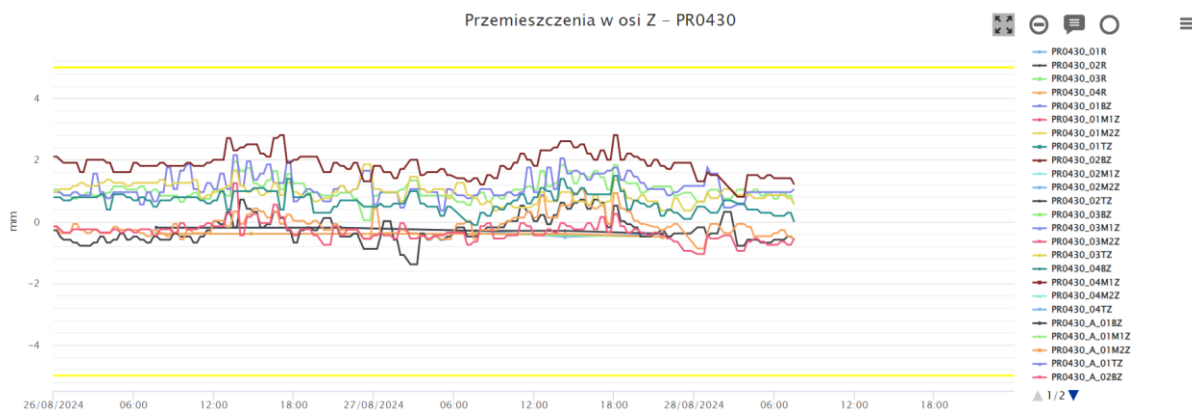


Figure 6-7 Settlement of building PR0430

On PR0460, monitoring points show a small downward trend.

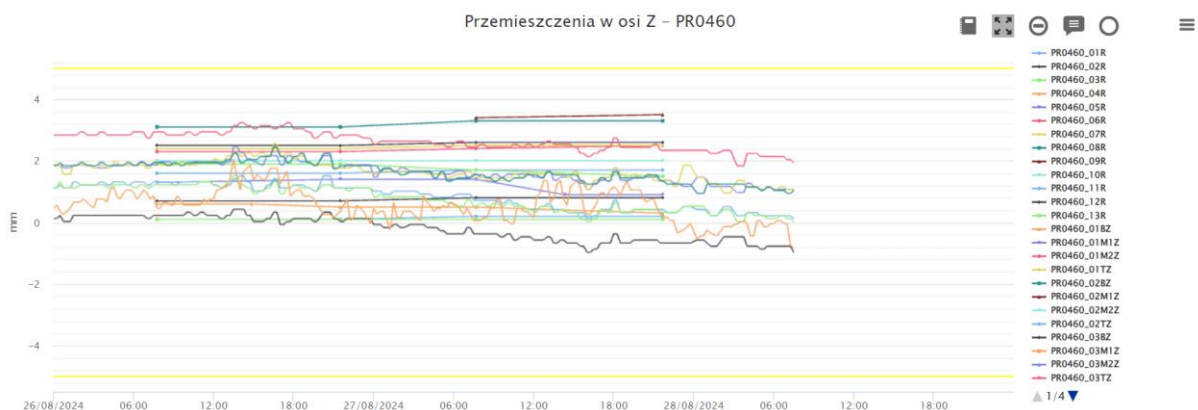


Figure 6-8 Settlement of building PR0460

On PR0440, monitoring points are stable.

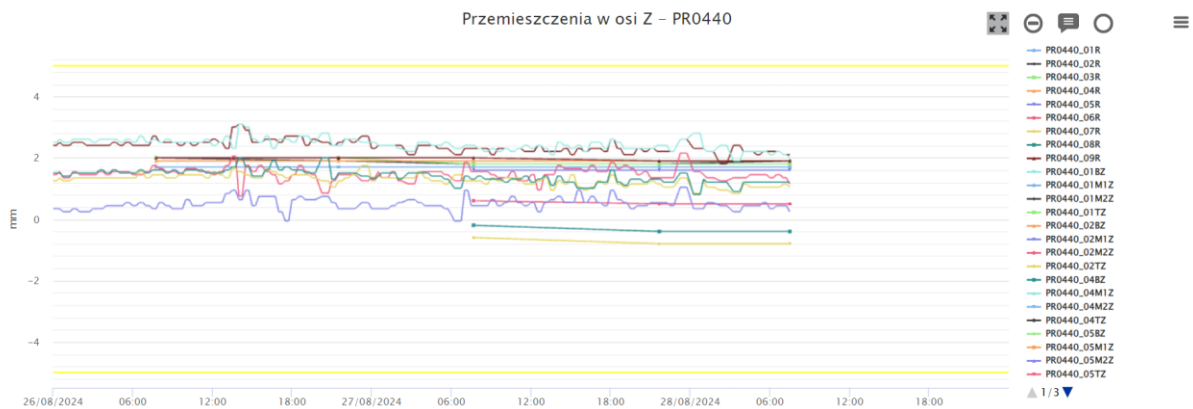


Figure 6-9 Settlement of building PR0440

7. COMPENSATION GROUTING ACTIVITIES

According to Raport Dobowy 27/08/2024 of Soletanche, compensation grouting has been performed on the 27/08/2024 from Shaft 2 in Próchnika 45.

The total amount of declared injected volume is 2040 l.

8. SOIL CONDITIONING

The volume of injected foam is plotted in the following Figure. The FIR value is computed based on the excavated material amount per ring and reported in the Table.

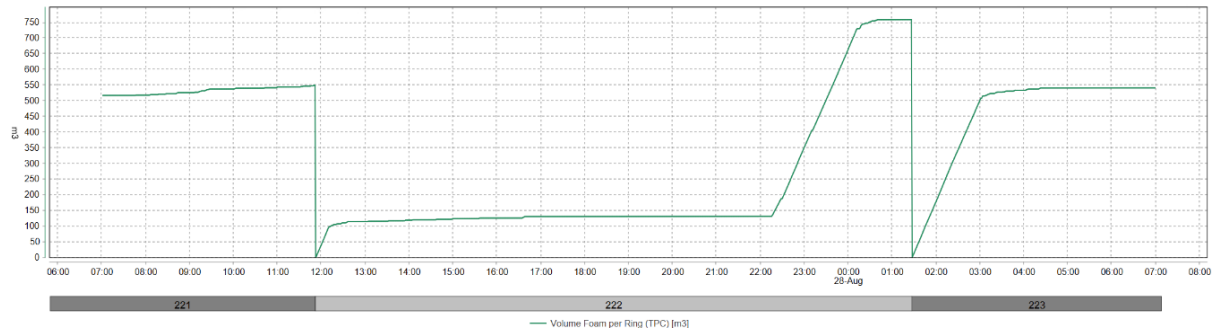


Figure 8-1 Injected foam volume

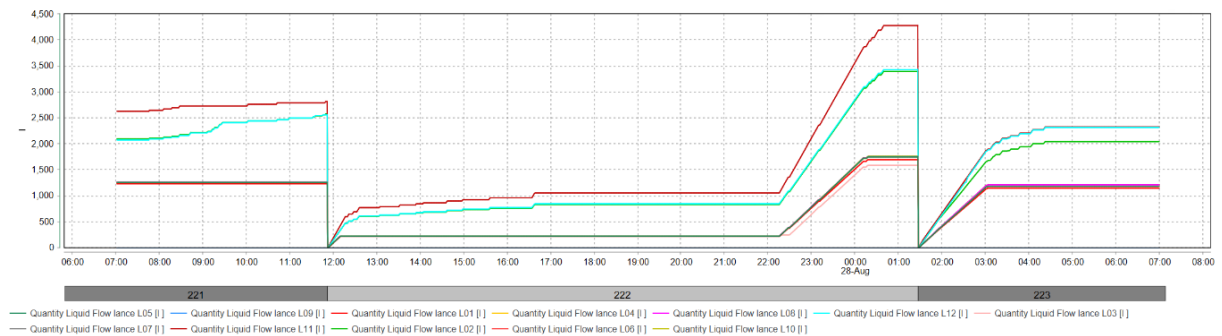


Figure 8-2 Conditioning liquid volume

Table 8-1 Soil conditioning data

Ring	Liquid volume (m³)	C (%)	FIR (%)
221	18.1	2	259
222	25	2	358